FLATNESS OF A QUASI-COHERENT MODULE AT A POINT OF AN ALGEBRAIC SPACE TESTED BY PRESENTATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof fixes one geometric point at the start and works with it throughout; the quantified statement over 'any geometric point lying over x' follows because the chosen one is arbitrary.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting geometric points along a surjective etale morphism. Properties of Spaces, Lemma spaces-properties-lemma-geometric-lift-to-usual, used at proof lines 6356-6360. Produces ubar over u with a o ubar = xbar.
- Etale local rings as strict henselizations of local rings of schemes. Properties of Spaces, Lemma spaces-properties-lemma-describe-etale-local-ring, used at proof lines 6361-6370. Identifies the two local rings appearing in the statement.
- Stalks of quasi-coherent sheaves described through an etale presentation. Properties of Spaces, Lemma spaces-properties-lemma-stalk-quasi-coherent, used at proof lines 6380-6387. Writes F_xbar as (a^*F)_u tensored with O_{X,xbar}.
- Flatness descends in the presence of the tensor description. Algebra, Lemma algebra-lemma-flatness-descends-more-general, used at proof lines 6388-6391. Equates flatness of (a^*F)_u over O_{V,v} with flatness of F_xbar.
- The flat up-and-down property of henselization. More on Flatness, Lemma flat-lemma-flat-up-down-henselization, used at proof lines 6392-6393. Replaces O_{V,v} by its henselization O_{Y,ybar} in the flatness statement.

Source reference. Original lemma 0584: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6767–6767. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7555–7555. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04KF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5971–5971. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7557–7557. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7558–7558. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $x \in |X|$. The following are equivalent*

(1) *for some commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes, a, b are étale, and $u \in U$ mapping to x the module $a^*\mathcal{F}$ is flat at u over V,

(2) *the stalk $\mathcal{F}_{\overline{x}}$ is flat over the étale local ring $\mathcal{O}_{Y,\overline{y}}$ where $\overline{x}$ is any geometric point lying over x and $\overline{y} = f \circ \overline{x}$.*

Proof. During this proof we fix a geometric proof $\overline{x} : \mathrm{Spec}(k) \to X$ over x and we denote $\overline{y} = f \circ \overline{x}$ its image in Y. Given a diagram as in (1) we can find a geometric point $\overline{u} : \mathrm{Spec}(k) \to U$ lying over u with $\overline{x} = a \circ \overline{u}$, see Properties of Spaces, Lemma 05VN. Set $\overline{v} = h \circ \overline{u}$ with image $v \in V$. We know that

$$\mathcal{O}_{X,\overline{x}} = \mathcal{O}^{sh}_{U,u} \quad \text{and} \quad \mathcal{O}_{Y,\overline{y}} = \mathcal{O}^{sh}_{V,v}$$

see Properties of Spaces, Lemma 04KF. We obtain a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_{U,u} & \longrightarrow & \mathcal{O}_{X,\overline{x}} \\ \uparrow & & \uparrow \\ \mathcal{O}_{V,v} & \longrightarrow & \mathcal{O}_{Y,\overline{y}} \end{array}$$

of local rings. Finally, we have

$$\mathcal{F}_{\overline{x}} = (a^*\mathcal{F})_u \otimes_{\mathcal{O}_{U,u}} \mathcal{O}_{X,\overline{x}}$$

by Properties of Spaces, Lemma 05VP. Thus Algebra, Lemma 0584 tells us $(a^*\mathcal{F})_u$ is flat over $\mathcal{O}_{V,v}$ if and only if $\mathcal{F}_{\overline{x}}$ is flat over $\mathcal{O}_{V,v}$. Hence the result follows from More on Flatness, Lemma 05VL. □